3.3 Proofs with Parallel Lines

Name _____ Period ____ Date _____

Geometry 3.3 (30-minute plan) · Student sheet

I can…

- use the Corresponding Angles Converse to decide whether lines are parallel.
- use the other converses to decide whether lines are parallel.
- use the Transitive Property of Parallel Lines.

Vocabulary · fill in the missing word

Converse — the statement formed by switching the _____ and the conclusion

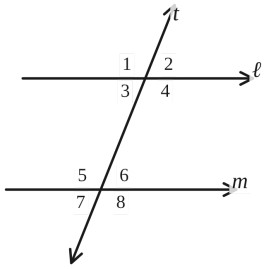
Transitive Property of Parallel Lines — _____

Part 1 · Notes and examples (fill in each note, then work the example)

Corresponding Angles Converse: if two lines are cut by a transversal so that corresponding angles are congruent, then the lines are _____.

Example 1

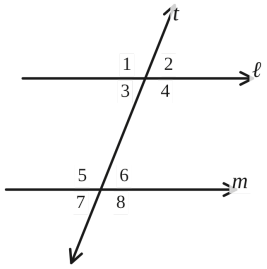
Find the value of x that makes $\ell \parallel m$: $m\angle 1 = (3x + 5)°$ and $m\angle 5 = 65°$.



Other converses: if alternate interior angles or alternate exterior angles are congruent, the lines are parallel; if consecutive interior angles are _____, the lines are parallel.

Example 2

$m\angle 4 = 75°$ and $m\angle 6 = 105°$. Are ℓ and m parallel? Explain.



Transitive Property of Parallel Lines: if two lines are parallel to the same line, then they are _____ to each other.

Example 3

Each rung of a ladder is parallel to the top rung. Are the rungs parallel to each other? Explain.

Part 2 · Partner practice

Partner practice

m∠4 = 112° and m∠6 = 68°. Are ℓ and m parallel? Explain.

